EE3032 Introduction to VLSI Design—Final

1. (15%) Figure 1 shows a circuit for realizing a low-cost latch. Complete the following questions.
   (a) What type of latch is realized by the circuit? (5%)
   (b) Assume that the switching threshold voltage of the inverter Inv1 is $V_{dd}/2$. List two inequalities for determining the sizes of P2 and N2. (10%)

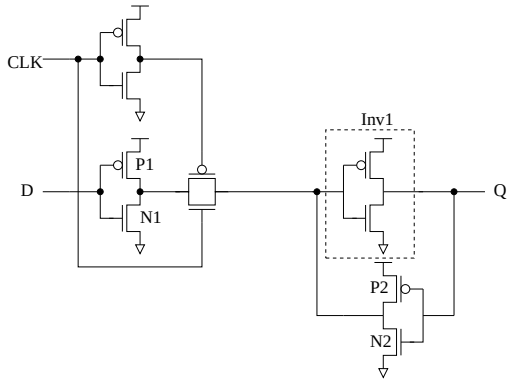

Figure 1: A low-cost latch.

2. (15%) Explain the operation principle of the circuit shown in Figure 2 shows a circuit for realizing a dynamic register. Answer the following questions.
   (a) Explain the operation principle of the circuit. (5%)
   (b) Design the register using the static low-cost latches shown in Figure 1. (10%)

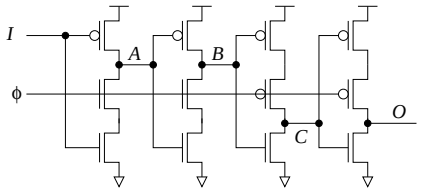

Figure 2: A dynamic CMOS register circuit.

3. (10%) Complete the following questions.
   (a) Figure 3(a) shows a three-input dynamic logic gate. Assume that the sizes of the NMOS and PMOS transistors of the dynamic logic gate and the inverter are equal and $C_g/C_d = \alpha$ (C_d=C_s). Also, the threshold voltage of the inverter is $V_{dd}/2$. Derive the minimal value of α such that the circuit can work correctly. (5%)
   (b) Complete the restore circuit of the Figure 3(b), such that Y=0 when X=0 and C=0. (5%)

4. (15%) Figure 4 shows a circuit designed with NP-domino logic. Complete the following questions.
   (a) What function does this circuit realize? (5%)
   (b) Using timing diagrams, describe the operation of the circuit. (5%)
   (c) Design the circuit using Domino logic (5%).

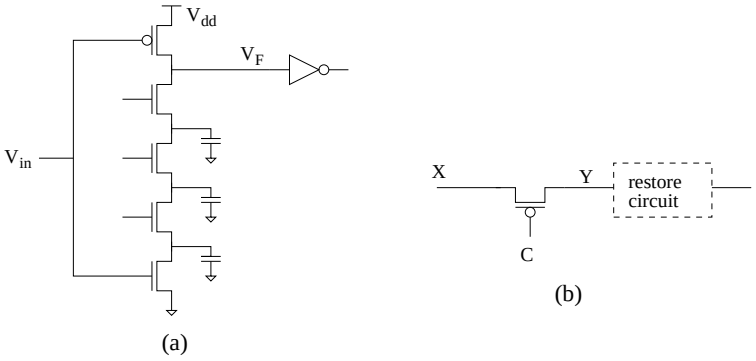

Figure 3: (a) dynamic logic (b) pass-transistor design.

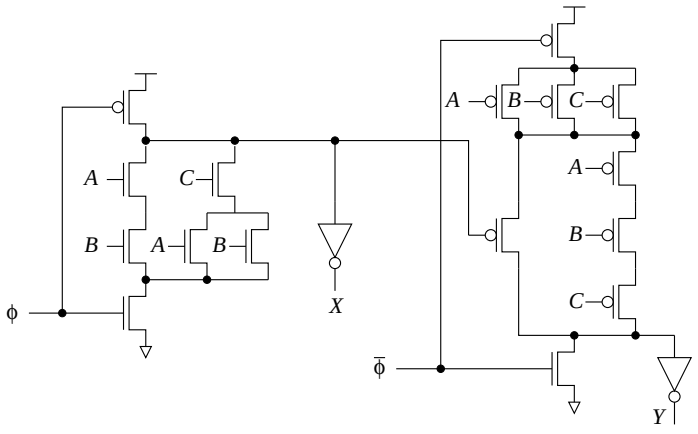

Figure 4: A circuit designed with NP-domino logic.

5. (10%) Figure 5 shows a sequential circuit (pipelined circuit). Assume that your boss asks you to design the circuit such that the circuit can be operated at 200MHz. Also, assume that the clock-to-Q propagation delay, the clock-to-Q contamination delay, the setup time, and the hold time of the registers R1 and R2 are $0.5ns$, $0.2ns$, $1.5ns$, and $1ns$, respectively. Determine the maximum and minimum delays of the combinational circuit.

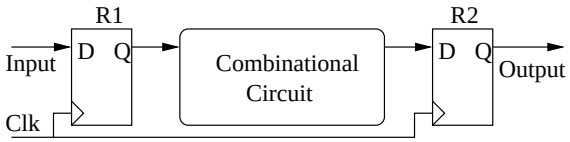

Figure 5: A pipelined circuit.

6. (10%) Figure 6 shows the symbolic layout of a circuit. Draw the transistor-level schematic diagram of the circuit.

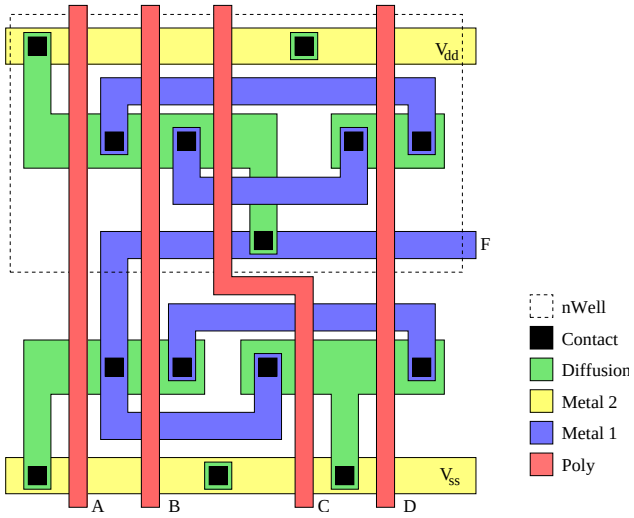

Figure 6: The symbolic layout.

7. (15%) Sketch the boolean function $f = \overline{AB + C + DE}$ using each of the following circuit techniques. (1) Domino logic (5%) (2) Fully differential logic (5%) (3) Static CMOS logic (5%)